

Open Source Solutions that Accelerate Adoption of Cognitive Automation

The concept of AI-Enabled Coworker is relatively new for many of the business enterprises.

If used effectively, it promises to transform the way we operate, work or live. It has the potential to make many business processes faster, simpler and less error-prone. In light of the heavy investments involved, this article directs readers to the open source solutions available, to accelerate the pace of its adoption.

The modern enterprise evolved over the late nineteenth and early twentieth centuries. Since then, technology has always been performing the role of a key enabler to expedite business innovation. Cognitive computing technologies seem to be yet another promising catalyst for enterprise transformation. These technologies aim at bringing about unprecedented levels of automation, and are poised to improve productivity across functions.

Cognitive systems attempt to simulate how humans think and learn. These systems imitate humans in learning from past experiences and use that knowledge for reasoning, making hypotheses, inferring, solving problems or making decisions. Combined with automation, enterprises can leverage these systems to automate even judgement based activities that are a part of a business process. This way, they can augment human skill and expertise so that our time is used more effectively.

Smart machines or *AI-enabled coworkers*, as analysts fondly call these systems, can simplify and accelerate many business processes to a great extent, and facilitate business agility and innovation. They can help a marketing executive by analysing the customer base and identifying the right target segment for the next campaign. They may indicate probable customer churn and help to identify the key parameters causing it. They can also help a service engineer by providing a note of caution regarding a potential outage. Or they may assist a call centre executive in reducing the resolution time for a query. They have the potential to transform the way the business operates today.

The pillars of cognitive technologies

The evolution of cognitive technologies involves various streams of artificial intelligence (AI). Many cognitive

technologies may be relevant for enterprises, including robotics, rules-based systems, computer vision, optimisation, planning and scheduling. However, in 2016 we expect the most important cognitive technologies in the enterprise software market to be:

- **Machine learning** – This is the ability of computer systems to improve their performance by exposure to data but without the need to follow explicitly-programmed instructions. This is likely to be the most prevalent. It enhances a large array of applications, from classification to prediction, and from anomaly detection to personalisation.

▪ **Natural language processing (NLP)** – This technology enables computers to process text in the same way as humans, for example, extracting meaning from text or even generating text that is readable, stylistically natural and grammatically correct. It has many valuable applications when incorporated in software that analyses unstructured text.

▪ **Speech recognition** – This is the ability to automatically and accurately transcribe human speech, and the technology that enables this is useful for applications that may benefit from

hands-free modes of operation.



Enterprise adoption

Many leading software companies have already discovered the potential of cognitive technologies to improve the core functionality of their products, generate new and valuable insights for customers, and improve business operations through automation. These benefits are simply too compelling for software companies to ignore.

Analysts predict that by the end of 2016, a majority of the world's largest enterprise software companies (by revenue), will have integrated cognitive technologies into their